

Assignment_Day4

1.Bank Account Management System

You are tasked with creating a simple bank account management system in Python. Implement a class called BankAccount with the following specifications:

The class should have private instance variables for account number, account holder name, and balance.

Include a constructor to initialize these variables.

Implement getter and setter methods for each instance variable to ensure encapsulation.

Implement methods to deposit and withdraw money from the account.

Ensure that the withdraw method checks if the account has sufficient balance before allowing withdrawal.

Write a Python program to demonstrate the functionality of the BankAccount class by creating instances, depositing and withdrawing money, and displaying account information.

Code:

```
class BankAccount:
```

```
    # Constructor
```

```
    def __init__(self, account_number, account_holder_name, balance):
```

```
        self.__account_number = account_number
```

```
        self.__account_holder_name = account_holder_name
```

```
        self.__balance = balance
```

```
    # Getters
```

```
    def get_account_number(self):
```

```
        return self.__account_number
```

```
    def get_account_holder_name(self):
```

```
        return self.__account_holder_name
```

```
def get_balance(self):  
    return self.__balance  
  
# Setters  
def set_account_holder_name(self, name):  
    self.__account_holder_name = name  
  
# Deposit  
def deposit(self, amount):  
    if amount > 0:  
        self.__balance += amount  
        print("Deposited:", amount)  
    else:  
        print("Invalid amount")  
  
# Withdraw  
def withdraw(self, amount):  
    if amount <= 0:  
        print("Invalid amount")  
    elif amount > self.__balance:  
        print("Insufficient balance")  
    else:  
        self.__balance -= amount  
        print("Withdrawn:", amount)  
  
# Display info  
def display_info(self):  
    print("\n--- Account Details ---")
```

```
print("Account Number:", self.__account_number)
print("Account Holder:", self.__account_holder_name)
print("Balance:", self.__balance)
```

```
# ----- Main Prog -----
```

```
print("=== Bank Account System ===")
```

```
acc_no = int(input("Enter account number: "))
name = input("Enter account holder name: ")
balance = float(input("Enter initial balance: "))
```

```
account = BankAccount(acc_no, name, balance)
```

```
while True:
```

```
    print("\n1. Deposit")
    print("2. Withdraw")
    print("3. Show Details")
    print("4. Change Name")
    print("5. Exit")
```

```
    choice = input("Enter your choice: ")
```

```
    if choice == "1":
```

```
        amt = float(input("Enter amount to deposit: "))
        account.deposit(amt)
```

```
    elif choice == "2":
```

```
    amt = float(input("Enter amount to withdraw: "))
    account.withdraw(amt)

elif choice == "3":
    account.display_info()

elif choice == "4":
    new_name = input("Enter new name: ")
    account.set_account_holder_name(new_name)
    print("Name updated")

elif choice == "5":
    print("Exiting... Thank you!")
    break

else:
    print("Invalid choice, try again")
```

Output:

=== Bank Account System ===

Enter account number: 123456

Enter account holder name: Amman

Enter initial balance: 2000

1. Deposit
2. Withdraw
3. Show Details
4. Change Name
5. Exit

Enter your choice: 2

Enter amount to withdraw: 100

Withdrawn: 100.0

1. Deposit

2. Withdraw

3. Show Details

4. Change Name

5. Exit

Enter your choice: 3

--- Account Details ---

Account Number: 123456

Account Holder: Amman

Balance: 1900.0

1. Deposit

2. Withdraw

3. Show Details

4. Change Name

5. Exit

Enter your choice: 2

Enter amount to withdraw: 2000

Insufficient balance

1. Deposit

2. Withdraw

3. Show Details

4. Change Name

5. Exit

Enter your choice: 5

Exiting... Thank you!

2. Employee Management System

You are developing an employee management system for a company. Implement a class called Employee with the following specifications:

The class should have private instance variables for employee ID, name, and salary.

Include a constructor to initialize these variables.

Implement getter and setter methods for each instance variable.

Implement a method to display employee information.

Implement a method to give a salary hike to an employee. The method should accept a percentage value by which the salary will be increased.

Write a Python program to demonstrate the functionality of the Employee class by creating instances, displaying employee information, giving salary hikes, and displaying updated employee information.

Code:

```
class Employee:
```

```
    def __init__(self, eid, emp_name, emp_salary):
```

```
        self.__eid = eid
```

```
        self.__emp_name = emp_name
```

```
        self.__emp_salary = emp_salary
```

```
    # getters
```

```
    def get_details(self):
```

```

        return self.__eid, self.__emp_name, self.__emp_salary

# setters

def update_name(self, name):
    self.__emp_name = name

def update_salary(self, salary):
    if salary >= 0:
        self.__emp_salary = salary
    else:
        print("Salary cannot be negative")

# display

def show(self):
    print("\nEmployee ID:", self.__eid)
    print("Employee Name:", self.__emp_name)
    print("Monthly Salary:", self.__emp_salary)

# salary hike

def increase_salary(self, percent):
    if percent > 0:
        self.__emp_salary += (self.__emp_salary * percent / 100)
        print("Hike applied successfully")
        print("Updated Salary:", self.__emp_salary)
    else:
        print("Enter valid percentage")

# extra feature

def yearly_salary(self):

```

```

        print("Annual Salary:", self.__emp_salary * 12)

# ----- main prog -----

print("***** Employee System *****")

eid = int(input("Enter ID: "))
name = input("Enter Name: ")
salary = float(input("Enter Salary: "))

e1 = Employee(eid, name, salary)

while True:

    print("\nChoose option:")
    print("1 - View Details")
    print("2 - Increase Salary")
    print("3 - Annual Salary")
    print("4 - Change Name")
    print("5 - Exit")

    ch = input("Enter choice: ")

    if ch == "1":

        e1.show()

    elif ch == "2":

        p = float(input("Enter percentage: "))
        e1.increase_salary(p)

```



```
elif ch == "3":
    e1.yearly_salary()

elif ch == "4":
    new_name = input("Enter new name: ")
    e1.update_name(new_name)
    print("Name changed")

elif ch == "5":
    print("Program ended")
    break

else:
    print("Wrong input, try again")
```

Output:

***** Employee System *****

Enter ID: 12345

Enter Name: Amman

Enter Salary: 5000

Choose option:

1 - View Details

2 - Increase Salary

3 - Annual Salary

4 - Change Name

5 - Exit

Enter choice: 2

Enter percentage: 25

Hike applied successfully

Updated Salary: 6250.0

Choose option:

1 - View Details

2 - Increase Salary

3 - Annual Salary

4 - Change Name

5 - Exit

Enter choice: 4

Enter new name: Farooque

Name changed

Choose option:

1 - View Details

2 - Increase Salary

3 - Annual Salary

4 - Change Name

5 - Exit

Enter choice: 1

Employee ID: 12345

Employee Name: Farooque

Monthly Salary: 6250.0

Choose option:

1 - View Details

2 - Increase Salary

3 - Annual Salary

4 - Change Name

5 - Exit

Enter choice: 5

Program ended